

Why the Proton Is Irreducibly Tripartite: the color-singlet obstruction to bipartite qudit merges

Tessera — cobordism subsystem (#382 / #396)

Abstract

We attempted to build a proton by *composing bipartite merges*: relax a cobordism whose boundary is two color states ψ_A, ψ_B , read the emergent result ψ_{AB} , then feed it as a boundary input of a second merge with a third state ψ_C . The emergent result never reaches the color singlet — it stays color-charged ($|\sigma| \approx 0.67$) at every step. This note shows that this is *not* a numerical or topological defect of the construction but a representation-theoretic theorem: the color singlet of $SU(3)$ is the three-index alternating invariant ε_{ijk} , which is *absent* from the two-fold tensor product $\mathbf{3} \otimes \mathbf{3}$ and first appears in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. A baryon singlet built from fundamentals is therefore irreducibly an $n = 3$ object; no single bipartite fusion of two color triplets can produce it. The qudit generalization makes the statement sharp: for $SU(d)$ the baryon arity equals d , so a *qubit* ($d = 2$) baryon *is* bipartite (the spin singlet ε_{ij}), while a *qutrit* color baryon ($d = 3$) is not. The simplicial resolution is the trivalent junction W_{ABC} that carries ε_{ijk} directly.

1 The question

The merge primitive is a Lorentzian Regge cobordism W with prescribed boundary ∂W : we pin the boundary color states, relax the interior edge lengths to a stationary point of the dual Regge action subject to a realizability constraint, and read the emergent result off the relaxed geometry. Composition is by *cobordism*, not welding: the emergent result-state of one merge is the boundary input of the next.

Concretely we merged three color-neutral $q\bar{q}$ pairs $A = (1, -1, 0)$, $B = (1, 0, -1)$, $C = (0, 1, -1)$ pairwise:

$$\text{merge}(A, B) \rightarrow AB, \quad \text{merge}(AB, C) \rightarrow ABC.$$

With the exact period read-out (`residualForPeriods` for the pinned inputs, `cyclePeriods` for the emergent result block) we measure the emergent color charge $\sigma(\psi) = \sum_i \psi_i$:

$$|\sigma_{AB}| \approx 0.70, \quad |\sigma_{ABC}| \approx 0.67, \quad \text{target singlet } |\sigma| = 0.$$

The result is color-charged at every step. *Is this a bug or a theorem?* It is a theorem.

2 Color, invariant tensors, and singlets

A quark transforms in the fundamental $\mathbf{3}$ of $SU(3)_{\text{color}}$, an antiquark in $\bar{\mathbf{3}}$. A physical (observable) hadron must be *colorless*: a singlet, i.e. an invariant under the color group. Building hadrons is therefore the problem of constructing $SU(3)$ -invariant tensors out of copies of $\mathbf{3}$ and $\bar{\mathbf{3}}$.

Theorem 1 (First fundamental theorem of invariant theory for $SL(3)$). *Every $SU(3)$ -invariant tensor in copies of $\mathbf{3}$ and $\bar{\mathbf{3}}$ is a polynomial in the two elementary invariants*

$$\delta^i_j \quad (\text{the contraction } \mathbf{3} \otimes \bar{\mathbf{3}} \rightarrow \mathbf{1}), \quad \varepsilon_{ijk}, \varepsilon^{ijk} \quad (\text{the alternating } \Lambda^3).$$

The two invariants are exactly the two ways to make a colorless object:

- δ^i_j pairs a quark with an antiquark: the **meson** $\bar{q}_i q^i$. It is *bilinear* — a colorless meson is a two-body (bipartite) object.
- ε_{ijk} contracts *three* quarks: the **baryon** $\varepsilon_{ijk} q^i q^j q^k$ (the proton). It is *trilinear* and totally antisymmetric.

The decisive fact is what is *missing*: there is no invariant bilinear in two fundamentals.

Proposition 1 (No singlet at two bodies).

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}, \quad \text{and neither summand is the trivial representation } \mathbf{1}.$$

Hence the two-quark Hilbert space $\mathbf{3} \otimes \mathbf{3}$ contains no color singlet. Equivalently, the multiplicity of $\mathbf{1}$ in $\mathbf{3}^{\otimes 2}$ is 0.

The singlet first appears one body later:

Proposition 2 (The singlet is born at three bodies).

$$\mathbf{3}^{\otimes 2} = \mathbf{6} \oplus \bar{\mathbf{3}} \quad (\dim 9 = 6 + 3), \quad \mathbf{3}^{\otimes 3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1} \quad (\dim 27 = 10 + 8 + 8 + 1).$$

The trivial representation occurs with multiplicity 0, 0, 1 in $\mathbf{3}^{\otimes 1}, \mathbf{3}^{\otimes 2}, \mathbf{3}^{\otimes 3}$; the unique copy in $\mathbf{3}^{\otimes 3}$ is spanned by ε_{ijk} .

Proposition 1 is the heart of the matter: *the proton state is not an element of any two-body color Hilbert space*. A bipartite fusion $V_A \otimes V_B \rightarrow V_R$ of two color triplets can only land in $\mathbf{6} \oplus \bar{\mathbf{3}}$; its output spectrum simply does not contain the singlet to which we are trying to relax. Compare the colorless meson: it lives in $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$, which *does* contain $\mathbf{1}$ — which is exactly why a $q\bar{q}$ pair (our carriable input) is a legitimate two-body colorless object, while a proton is not.

3 The associativity subtlety: why a *sequence* of merges still fails

One might object that the singlet *is* reachable pairwise through associativity:

$$(\mathbf{3} \otimes \mathbf{3}) \otimes \mathbf{3} = (\mathbf{6} \oplus \bar{\mathbf{3}}) \otimes \mathbf{3} = \underbrace{(\mathbf{6} \otimes \mathbf{3})}_{\mathbf{10} \oplus \mathbf{8}} \oplus \underbrace{(\bar{\mathbf{3}} \otimes \mathbf{3})}_{\mathbf{1} \oplus \mathbf{8}}.$$

The singlet appears *only* in the $\bar{\mathbf{3}} \otimes \mathbf{3}$ channel: one must first project the diquark onto its *antisymmetric* part $\bar{\mathbf{3}} = \varepsilon_{ijk} q^j q^k$ and discard the symmetric sextet $\mathbf{6}$. A bipartite merge that *projects onto the antisymmetric channel* could then complete the singlet with the third quark.

But the emergent merge does no such projection. Relaxing the cobordism produces a stationary geometry carrying the *generic* product $\mathbf{6} \oplus \bar{\mathbf{3}}$; the symmetric $\mathbf{6}$ component is not annihilated, and it contaminates the second merge — the result stays colored ($|\sigma_{AB}| \approx 0.70$). To kill the $\mathbf{6}$ one must impose the *simultaneous* antisymmetry of all three color indices, ε_{ijk} , at a *single* vertex. That object is a trivalent junction carrying the alternating (sign) representation — not any composition of generic pairwise merges.

Remark 1. This is the precise sense in which the proton is “irreducibly tripartite”: the defining invariant $\varepsilon_{ijk} \in \Lambda^3(\mathbb{C}^3)^*$ is a totally antisymmetric *three*-form. It does not factor as a composite of bilinear invariants except by first manufacturing the antisymmetric diquark $\bar{\mathbf{3}}$ — and that projection is itself the statement that the three indices must meet antisymmetrically. There is no associative bracketing that turns a *generic* two-body fusion into a singlet.

4 The qudit generalization (the sharp answer)

The argument is not special to $SU(3)$; it pins the baryon *arity* exactly. For $SU(d)$ acting on a qudit (the fundamental $\mathbb{C}^d$), the invariant tensors are again generated by δ^i_j and the Levi-Civita symbol $\varepsilon_{i_1 \dots i_d}$, now with d indices.

Corollary 1 (Baryon arity = d). *A singlet built from fundamentals of $SU(d)$ requires exactly d of them: $\varepsilon_{i_1 \dots i_d} q^{i_1} \dots q^{i_d}$. The multiplicity of $\mathbf{1}$ in $\mathbf{d}^{\otimes n}$ is 0 for $0 < n < d$ and first becomes nonzero at $n = d$.*

This is the direct answer to “why can’t bipartite qubit/qudit interactions reach the proton”:

- **Qubit**, $d = 2$. ε_{ij} has *two* indices: $\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}$ already contains a singlet (the spin-0 state $\frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$). A “baryon” of qubits *is* bipartite. Bipartite merges suffice here — which is exactly why qubit intuition misleads.
- **Qutrit / color**, $d = 3$. ε_{ijk} has *three* indices: $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ has no singlet. The proton is a qutrit baryon and is irreducibly *tripartite*.

The proton’s three-quark structure is, in one line, the statement $d = 3$: the color singlet is the d -fold alternating invariant, and $d = 3$ for color.

5 The simplicial model and the measurement

In the cobordism subsystem the color register is the homology of the holed register surface: $b_1 = 2$ on the $\sigma = 0$ hyperplane, carrying the *standard* (2-dimensional) representation of $S_3 = \text{Weyl}(SU(3))$ permuting the three colors, with

$$\mathbb{C}^3 = \underbrace{\mathbf{1}}_{\sigma \neq 0 \text{ direction}} \oplus \underbrace{V_{\text{std}}}_{\sigma = 0 \text{ register}}.$$

Confinement is the topological statement that the $\sigma \neq 0$ (trivial-rep) direction is *not carried* by the register: a net color charge has no realizable harmonic representative, while a $\sigma = 0$ configuration does. The proton singlet is represented by the definite-triality state

$$\psi_{\text{proton}} = (1, \omega, \omega^2), \quad \omega = e^{2\pi i/3}, \quad 1 + \omega + \omega^2 = 0,$$

the $\mathbb{Z}_3 \subset S_3$ Fourier mode: it is the discrete shadow of the alternating invariant ε_{ijk} (an eigenvector of the cyclic color rotation, i.e. a state of definite triality, lying in the $\sigma = 0$ register).

A bipartite merge is a cobordism W with $\partial W = \psi_A \sqcup \psi_B$ and an emergent result block R ; we read $\sigma_R = \sum_i (\psi_R)_i$ from the relaxed metric via `cyclePeriods`. Consistent with Proposition 1, the emergent two-input result does not land in the singlet direction:

$$|\sigma_{AB}| \approx 0.70, \quad |\sigma_{ABC}| \approx 0.67 \quad (\text{sequence}),$$

against the singlet target $|\sigma| = 0$. Two control facts confirm the reading is faithful and the obstruction structural:

- The result is read with the *exact* period machinery (`residualForPeriods/cyclePeriods`), not the soft edge-loop encoding; the $\approx 10^{-2}$ neutral realizability floor is intrinsic to the shared-register staircase, not an artifact of the read-out. The colored/neutral confinement split (ratio $\sim 10^3$) is intact.
- Each merge step is a genuine valid manifold ($b_1 = 2$, every triangle in ≤ 2 tetrahedra), so the failure to reach the singlet is not a degeneration of the cobordism but a property of the two-body color content.

6 Resolution: the trivalent junction W_{ABC}

The representation theory dictates the fix. Since ε_{ijk} is a *three*-valent invariant, the proton must be realized by a cobordism with *three* boundary inputs meeting a single bulk,

$$\partial W_{ABC} = \psi_A \sqcup \psi_B \sqcup \psi_C,$$

a trivalent junction whose induced orientation on the three color cycles carries the alternating (sign) structure — the simplicial ε_{ijk} . This is the $n = 3$ object in which the singlet first occurs, built once rather than as a sequence of two-body fusions. The exact period machinery used here (`readoutHoles/residualForPeriods` to pin the three inputs, `cyclePeriods` to read the emergent result, `emergesResult` to pin inputs only) is exactly what that junction consumes; only the junction topology itself is new (tracked as #396).

7 Conclusion

The proton does not emerge from bipartite merges because the $SU(3)$ color singlet is the three-index alternating invariant ε_{ijk} , which is absent from $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and first appears in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. The two-quark color Hilbert space simply does not contain the state we were relaxing toward; a sequence of generic pairwise fusions cannot manufacture the simultaneous antisymmetry of three color indices. The qudit form of the statement is exact — baryon arity = d , so qubits make baryons bipartitely but qudit color does not — and our simplicial register reproduces it quantitatively: the emergent bipartite result stays color-charged ($|\sigma| \approx 0.67$) at every step. The proton is irreducibly tripartite, and its carrier is the trivalent junction W_{ABC} .

8 Addendum (#416): the twisted antisymmetrizer and the connected path

Sec. 3 conceded the singlet *is* reachable pairwise *if* the first merge projects onto the antisymmetric diquark $\bar{\mathbf{3}} = \varepsilon_{ijk} q^j q^k$. The claim there was that the *generic emergent* merge does no such projection. We now test both the projection and the deeper confinement-of-intermediate claim directly, measuring *through* the existing prism triangulation on the uniform $\ell^2 = 1$ metric (the seed-independent geometric transport; the jittered relax is a noisy proxy whose $|\sigma|$ ranges 0.24–1.02 across seeds).

(a) An orientation-reversing twist is an *exact* antisymmetrizer. Extruding the holed-icosahedron register tube with a twist that reverses each color hole’s induced orientation (a within-hole vertex transposition, supplied to `Spacetime::prismCells`’s cumulative twist via `RegisterTopology::setTwist` flips the sign of each carried color period. On the uniform metric the two input-transport blocks then satisfy $M_B = -M_A$ *exactly* ($\|M_A + M_B\| \approx 6 \times 10^{-15}$), so the symmetric sextet cancels: the $A \leftrightarrow B$ antisymmetric fraction of the emergent diquark jumps from 0.00 (untwisted control — the *pure* symmetric $\mathbf{6}$) to 1.00 (twisted — the *pure* antisymmetric $\bar{\mathbf{3}}$). The “generic” merge does not project, exactly as Sec. 3 says; but the projection is not unreachable — it is a single orientation-reversing twist, the simplicial ε_{ijk} acting on the two-body factor. (The untwisted control’s $|\sigma| \approx 0.70$ of line 161 is a jittered-relax artifact; the faithful uniform measurement shows the control is the *pure* sextet, antisymmetric fraction 0.)

(b) **Confinement of the intermediate is of the *free* diquark only.** The colored diquark $\bar{\mathbf{3}}$ is over-determined on the *shared* bipartite register (its realizability residual $r_U \approx 15$, the same averaging that floors every state there), so the welded/free-boundary sequence — read the diquark out as a free boundary and re-pin it — floors at singlet overlap ≈ 0.55 . But on *distinct* windows (independent cycles) the *same* colored state carries *exactly*, $r_U \approx 4 \times 10^{-27}$. A *connected* path that keeps the diquark an interior carried state on its own cycle (∂W = the input/result windows only, no free diquark boundary) therefore has a full-rank input→result transport: the singlet $[1, \omega, \omega^2]$ lies in its image (reachability 1.00), and charge is conserved at the junction ($|\sigma_R| \approx 9 \times 10^{-15}$ for neutral inputs).

Verdict. Bipartite-*with*-projection *does* produce the antisymmetric diquark $\bar{\mathbf{3}}$ (antisymmetric fraction $0 \rightarrow 1$ under the orientation-reversing twist), and the color singlet *is* reachable through a connected path that keeps the diquark interior on an independent cycle (reachability 1.00, against the welded free-boundary sequence’s 0.55). The confinement-of-the-intermediate is the obstruction to the *welded* sequence — the free colored diquark is non-carriable on a shared register — and it is *not* fundamental: keeping the intermediate interior ($r_U \rightarrow 0$ on a distinct cycle) escapes it. This refines rather than overturns the tripartite conclusion: a single two-body fusion still has no singlet (Prop. 1 is untouched), but a *two-step* bipartite construction with an explicit antisymmetrizing projection and an interior intermediate reaches it — which is the sequential image of the same ε_{ijk} the star junction W_{ABC} carries at one vertex.